Ans to the ques no-1

Instruction	Type	Binary-Hex
slli t0, x22, 3	I	00000000000111011000100101100011 0x00351293
add t1, x22, t0	R	00000000001011010100000011001100 0x005A8333
ld t2, 0(t1)	I	00000000000001100110011110000011 0x00033383
addi t3, t0, 16	I	00000001000000010100011110000100 0x01028e13
add t4, x21, t3	R	00000001110010101000111010110011 0x01CA8eB3
ld t5, 0(t4)	I	00000000000001110101111111000011 0x000EBF03
ssi t6, t5, 3	I	00000000000011111100011111001011 0x003F1F93
add t1, x20, t6	R	00000001111111010000000011001011 0x01FA0333
addi t4, t4, -1	I	00000000000000011001111100000001 0x00033E03
addi t5, t0, 8	I	00000000001111110000011101011001 0x00FE0CB3
add t6, x20, t5	R	00000001111101010000001111011001 0x01EA0FB3
sd t4, 0(t6)	S	
ld t3, 0(t1)	I	00000000000000011001111100000001 0x01FA0333
add t4, t3, t2	R	00000000000111111000001101011001 0x00FE0CB3

Ans to the ques no-2

```
mov x22, 5
mov s1, 1
```

P.T.O

```
Loop:
    blt  x22, s1, Exit
    slli t0, x22, 3
    add  t1, x21, t0
    ld   t2, 0(t1)
    ld   t3, 0(x20)
    beq  t2, t3, Else
    add  t4, x20, t0
    ld   t5, 0(t4)
    addi t5, t5, 4
    addi t6, x21, 16
    sd   t5, 0(t6)
    j    Endif
Else:
    beq  t2, t3, Inner then
    ld   t5, 0(t1)
    addi t5, t5, 4
    addi t6, x20, 16
    sd   t5, 0(t6)
    j    Endif
Inner than:
    addi t6, x20, 16
    ld   t5, 0(t6)
    add  t3, t5, 4
    sd   t5, 0(t1)
Endif
    addi x22, x22, -1
    j    loop
Exit:
```

P.T.O

Ans to the ques no-3

```
    mov  t0, 0
    mov  x22, 0
Loop:
    bge  x22, x10, Exit
    slli t1, x22, 3
    add  t2, x20, t1
    ld   t3, 0(t2)
    add  t4, x21, t1
    ld   t5, 0(t4)
    ble  t3, t5, Skip
    sub  t6, t3, t5
    add  t0, t0, t6

SKP:
    addi x22, x22, 1
    J    Loop
Exit: mov x10, t0
    ret
```

Ans to the. ques no 4

```
Weighted-sum-Strided
    mov t0, 0
    mov x22, 0
Loop:
    bge  x22, x10, Exit
    slli  t1, x22, 3
    add  t2, x20, t1
    ld   t3, 0(t2)
    add  t4, x21, t1
    ld   t5, 0(t4)
    addi  t6, t3, 1
    bne   t6, x0, next
    ble   t3, t5, next
    sub   t6, t3, t5
    mul , t6, t6, x11
    add   t0, t0, t6
Next :
    addi   x22, x22, 2
        J  loop
Exit .
    mov  x10 t0
    ret
```

Ans to the ques no-5

```
slli   t0, x22, 4
slli   t1, x22, 3
add    t0, t0, t1
add    t0, t0, x22
lwai   x21, t0, 4
```

Ans to the ques no-6

```
for (i=0, i<size; i++) {
    int val = A[i];
    if ((val & 01) == 01) {
        val = val + h;
        B[h] = val;
    }
}
```

Ans to the ques no-7

opcode = 1100011, funct3 = 100

rs1 = x5, rs2 = x10

3 bit signal = 101111111100 = -2052

Target address = 0x4010 + (0-2052) = 0x380c